## Arithmetic Progression

(i) Arithmetic Progression is $a,\ a+d,\ a+2d,\ a+3d,\ldots.$ $n^{th}$ term is given by $t_n = a+(n-1)d$

(ii) Sum to first $n$ terms of an A.P. is $S_n = \frac{n}{2}[2a+(n-1)d]$

(iii) If the last term $l$ ($n^{th}$ term) is given, then $S_n = \frac{n}{2}[a+l]$

## Geometric Progression

(i) Geometric Progression is $a,\ ar,\ ar^2,\ldots,ar^{n-1}$. $n^{th}$ term is given by $t_n = ar^{n-1}$

(ii) Sum to first $n$ terms of an G.P. is $S_n = \frac{a(r^n-1)}{r-1}$ if $r \neq 1$

(iii) Suppose $r=1$ then $S_n = na$

(iv) Sum to infinite terms of a G.P. $a+ar+ar^2+\cdots$ is $S_\infty = \frac{a}{1-r}$, where $-1 < r < 1$

## Special Series

(i) The sum of first $n$ natural numbers $1+2+3+\cdots+n = \frac{n(n+1)}{2}$

(ii) The sum of squares of first $n$ natural numbers

$1^2+2^2+3^2+\cdots+n^2 = \frac{n(n+1)(2n+1)}{6}$

(iii) The sum of cubes of first $n$ natural numbers $1^3+2^3+3^3+\cdots+n^3 = \left[\frac{n(n+1)}{2}\right]^2$

(iv) The sum of first $n$ odd natural numbers $1+3+5+\cdots+(2n-1) = n^2$

## ICT CORNER

### ICT 2.1

**Step 1:** Open the Browser type the URL Link given below (or) Scan the QR Code. GeoGebra work book named **"Numbers and Sequences"** will open. In the left side of the work book there are many activity related to mensuration chapter. Select the work sheet **"Euclid's Lemma division"**

**Step 2:** In the given worksheet Drag the point mentioned as **"Drag Me"** to get new set of points. Now compare the Division algorithm you learned from textbook.

Step 1



Step 2



Expected results



### ICT 2.2

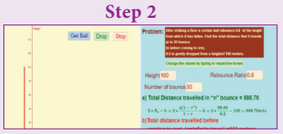
**Step 1:** Open the Browser type the URL Link given below (or) Scan the QR Code. GeoGebra work book named **"Numbers and Sequences"** will open. In the left side of the work book there are many activity related to mensuration chapter. Select the work sheet **"Bouncing Ball Problem".**

**Step 2:** In the given worksheet you can change the height, Number of bounces and debounce ratio by typing new value. Then click **"Get Ball"**, and then click **"Drop"**. The ball bounces as per your value entered. Observe the working given on right hand side to learn the sum of sequence.

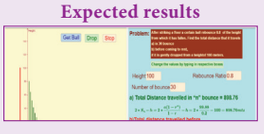
Step 1



Step 2



Expected results



*You can repeat the same steps for other activities*

https://www.geogebra.org/m/jfr2zzgy#chapter/356192

or Scan the QR Code.

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